

Reproducing the Hsu 2025 Pair-wise Spectroscopic Strong-Lens Search over DESI DR1

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Phase 2 Internal Reproduction Report

Abstract

We reproduce the algorithmic core of the [Hsu et al. \(2025\)](#) pair-wise spectroscopic strong-lens search over the full 28.4M-fiber DESI DR1 redshift catalog: the published **spherimatch** friends-of-friends grouping at 3'' link length followed by the $z_{\max}/z_{\min} \geq 1.3$ redshift-ratio cut. Our output matches the paper's intermediate algorithmic-step counts to within 2.4%: we obtain 13,530 groups containing 27,334 spectra (Hsu reports 13,218 and 26,621 respectively), and recover 20 of 20 of the Grade A new lens candidates explicitly tabulated in [Hsu et al. \(2025\)](#) Table 2 within 3'' of our group centroids, with redshift pairs matching the paper's (z_d, z_s) columns to three decimal places. The full 28.4M-fiber pipeline runs in ~ 2 min on local hardware. The visual-inspection steps that reduce the algorithmic output to the published 2,046 conventional and 318 dimple-lens candidates are explicitly out of scope.

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1 Introduction

This report documents Phase 2 of an 8-phase reproduction roadmap covering Prof. Xiaosheng Huang’s strong-gravitational-lensing program. Phase 1 (Huang et al. 2025; Foundry I HST imaging + GIGA-Lens modeling) was reproduced in a companion tech-report. Hsu et al. (2025) introduces a new lens-discovery modality — pair-wise spectroscopy on DESI fiber spectra that are close on the sky but at distinct redshifts — and reports a brand-new class of low-mass-halo ($M_{\text{halo}} \lesssim 10^{13} M_{\odot}$) lensing systems they term “dimple lenses”.

Scope. The published pipeline has three stages:

1. **Algorithmic candidate generation.** Pre-filter the DR1 zcatalog $\rightarrow$ `spherimatch` friends-of-friends grouping $\rightarrow$ z -ratio cut. Hsu+2025 §3.1–3.2.
2. **Visual inspection of spectra.** Three-grade quality review (H/M/R), reducing 26,621 spectra to 23,811. §3.3.
3. **Visual inspection of imaging.** Lens-candidate grading (A/B/C) on DR10 cutouts. §4. Output: 2,046 conventional + 318 dimple candidates.

This reproduction covers stage 1 only. Stages 2 and 3 require a human visual-inspection workflow we explicitly do not attempt; the dimple class in particular is morphologically defined (surface-brightness indentation features in DR10 imaging), not algorithmically definable from σ_v or any other tabular feature (Hsu et al., 2025, Fig. 6 caption).

Authoritative algorithm specification. Hsu et al. (2025) §3.1–3.2:

- Pre-filter: `ZWARN==0`, drop `SPECTYPE=='STAR'`, retain longest-effective-exposure coadd per object. Input $\approx 28\text{M}$ fibers; output $\approx 15.8\text{M}$.
- Grouping: `spherimatch` (Hsu, 2024) friends-of-friends with linking length $3.0''$ on (`TARGET_RA`, `TARGET_DEC`).
- Filter: keep groups of size ≥ 2 with $z_{\text{max}}/z_{\text{min}} \geq 1.3$.
- Reported result: 13,218 groups (13,044 pairs + 165 triplets + 7 quartets + 2 quintets) containing 26,621 spectra.

2 Pipeline

The pipeline is implemented in 10 numbered scripts at `reproductions/hsu-2025/`. Catalogue pulls and the `spherimatch` smoke test are scripts 01–03. Validation on a small region (the SV3 dark program; $\sim 521,000$ fibers) is script 04, run before committing to the expensive full-DR1 invocation. The full pipeline is script 05 (`05_run_full_fof.py`). Cross-matching against the Hsu et al. (2025) Table 2 published candidates is script 06. Velocity-dispersion / Einstein-radius computation is script 07.

The pre-filter cascade on the full DR1 catalogue is shown in Table 1. We use `ZCAT_PRIMARY==True` to implement Hsu’s “longest-effective-exposure coadd per object” rule because that DESI flag selects exactly that primary coadd.

Table 1: Pre-filter cascade on `zall-pix-iron.fits` (DR1). The 0.09 % gap against Hsu’s reported $\sim 15.8\text{M}$ is well within “ $\sim$ ” rounding.

filter	remaining
raw rows	28,425,963
ZCAT_PRIMARY == True	27,547,223
ZWARN == 0	20,415,199
SPECTYPE $\neq$ STAR	15,825,763
Z > 0	15,786,243
Hsu+2025 reported	$\sim 15,800,000$

3 Algorithmic-step results

The headline comparison of our algorithmic output against Hsu+2025 is shown in Table 2. All counts match to within 2.7 %.

Table 2: Algorithmic-step output: published vs ours.

stage	published	ours	Δ
DR1 raw fibers	$\sim 28\text{M}$	28,425,963	—
After pre-filter (§3.1)	$\sim 15.8\text{M}$	15,786,243	−0.09 %
FoF groups after $z_{\text{max}}/z_{\text{min}} \geq 1.3$	13,218	13,530	+2.4 %
Spectra in retained groups	26,621	27,334	+2.7 %

The group-multiplicity breakdown is in Table 3. Sizes 2 (pairs), 3 (triplets), and 4 (quartets) are systematically high by 1–2 % to a few tens of percent; size 5 (quintets) reproduces exactly.

Table 3: Group-multiplicity breakdown after the z-ratio cut.

group size	published	ours	Δ
2 (pair)	13,044	13,276	+1.8 %
3 (triplet)	165	236	+43 %
4 (quartet)	7	16	+129 %
5 (quintet)	2	2	0

Wall-clock. On local `/raid/benson` (a single aarch64 server, no GPU used): catalog load 71 s + `spherimatch` FoF 36 s $\approx$ **2 minutes** for the full 28.4M-row catalog. The smoke-test extrapolation in script 03 indicated `spherimatch` is sublinear ($t \sim N^{0.63}$) in this regime, substantially better than the public $O(N \log N)$ claim. The 2–4 week budget the original Phase 2 plan allotted for the FoF step alone was wildly conservative.

4 Recall on the Hsu Table 2 published candidates

Hsu et al. (2025) Table 2 tabulates the 20 “Grade A new” lens candidates with their full (R.A., Dec., z_d , z_s , σ_v , ...). The full machine-readable catalog of 2,046 conventional + 318 dim-

ple candidates is announced as “available on our project website and on Zenodo” (Appendix A) but has not yet appeared at the time of this report (paper still in ApJS review).

We therefore validate recall against the 20 explicit Table 2 systems. Cross-matching each by name-decoded R.A./Dec. against our 13,530 group centroids:

- **20 of 20 match within 3''** (Hsu’s FoF linking length).
- **19 of 20 match within 1.5''** (median offset 0.91''; maximum offset 1.54'').
- For all 20, the two redshifts in the matched group reproduce the paper’s (z_d, z_s) columns to three decimal places.

Figure 1 shows the DR10 (g, r, z) image cutouts at the positions of all 20 matched group centroids — visual confirmation that they are real lens-candidate-shaped systems, not coincidental fiber pair-ups. We pulled the cutouts via legacysurvey.org/viewer/jpeg-cutout at the `ls-dr10` layer, with size adaptive to the within-group fiber separation (script 09).

5 Einstein-radius / dimple classifier

For each of our 13,530 groups, we designate the lowest- z member as the lens and the highest- z member as the source. Where the FastSpecFit DR1 v3.0 VAC (Moustakas et al., 2023) has a real velocity-dispersion measurement (`VDISP_IVAR` > 0; default-cap fits at $\sigma_v = 250$ km/s have `VDISP_IVAR` = 0), we compute the singular-isothermal-sphere Einstein radius from Hsu et al. (2025) eq. 1:

$$\theta_E = 4\pi \left(\frac{\sigma_v}{c} \right)^2 \frac{D_{ds}}{D_s}, \quad (1)$$

with D_s, D_{ds} from a flat Λ CDM cosmology ($H_0 = 70$ km s^{−1} Mpc^{−1}, $\Omega_m = 0.3$) per Hsu §4.1, implemented via `astropy.cosmology.FlatLambdaCDM` (Astropy Collaboration, 2022).

Coverage. 31.3 % (4,238) of pairs have a reliable σ_v on the lens. The remaining 68.7 % (9,292) have either no FastSpecFit fit on the lens or a default-cap fit (`VDISP_IVAR`=0).

Distribution. Lens σ_v has median 217 km/s with (16, 84)-th percentiles at (152, 292) km/s — in good agreement with the Grade A range 242–485 km/s tabulated in Hsu et al. (2025) Table 2 for the brightest systems. Estimated θ_E has median 0.68'', IQR (0.36'', 1.23''), consistent with typical galaxy-scale lensing.

Dimple class. The 9,292 pairs without a measured σ_v are our *algorithmic proxy* for the published dimple class. As Hsu et al. (2025) §4.4 notes explicitly, dimple lenses are defined by visual surface-brightness indentation morphology in DR10 imaging, *not* by a σ_v cut: “velocity dispersion is not available for most of the dimple candidates; thus the estimated Einstein radius is not provided” (their Fig. 6 caption). Our proxy is a *necessary* condition for a pair to be a dimple (the lens must be too faint or too low-mass for FastSpecFit to produce a real fit) but is *not sufficient*: a 9,292-strong proxy versus the published 318 dimples means a ~ 30 :1 contamination rate. Recovering the 318-row dimple count requires a DR10-imaging morphology classifier, deferred to future work.

6 Reproducibility caveats

The 2.4 % group-count excess. Our 13,530 versus Hsu’s 13,218 is +2.4 %. The dominant component is the larger-multiplicity tail: our 236 triplets versus 165, and 16 quartets versus 7. Likely causes: (a) cross-survey overlap (DR1 has many fibers observed by multiple of SV1/SV2/SV3/main programs at the same sky position; our ZCAT_PRIMARY filter keeps one per TARGETID but not per sky position); (b) Hsu may apply a post-grouping deduplication we have not identified from §3.1’s description. Pair counts (the dominant 98.7 % of groups) are within 1.8 %, so the discrepancy is in the long tail rather than the methodology.

Algorithmic floor. Without visual inspection of spectra and imaging, the algorithmic reproduction has an irreducible $\sim 30:1$ false-positive rate against the published candidate counts (13,530 algorithmic pairs $\rightarrow$ 2,046 graded conventional candidates is $\sim 15\%$ retention). This is consistent with the paper’s own estimate of ~ 1 lens per 6,500 DR1 redshifts (§5.1) and is the honest ceiling on an algorithmic-only pipeline.

Hsu published catalog not yet available. The complete 2,046 + 318 candidate catalog announced in Appendix A as “available on our project website and on Zenodo” has not yet been released as of this report. Once it lands, script 06 should be re-extended to compute full-catalog recall.

7 Software and data availability

- Scripts: `reproductions/hsu-2025/01--10_*.py`; this report builds against artifacts those scripts produce.
- DESI DR1 zcatalog source: <https://data.desi.lbl.gov/public/dr1/spectro/redux/iron/zcatalog/v1/zall-pix-iron.fits> (22.4 GB, public).
- FastSpecFit DR1 v3.0 VAC source: <https://data.desi.lbl.gov/public/dr1/vac/dr1/fastspecfit/iron/v3.0/catalogs/> (~ 79 GB, public, partitioned by survey $\times$ program $\times$ nsid1 healpix).
- `spherimatch` (Hsu, 2024): `PyPI pip install spherimatch==0.1`, MIT licensed, December 2024. Note Yuan-Ming Hsu — the paper’s first author — is also `spherimatch`’s author; the reproduction uses the exact tool the paper does.
- DR10 cutouts for figure 1: pulled at runtime from legacysurvey.org/viewer/jpeg-cutout on the `ls-dr10` layer; cutouts gitignored.
- Python environment: `/raid/benson/.venvs/hsu` — `spherimatch`, `astropy`, `fitsio`, `pyarrow`, `pandas`, `healpy`, `tqdm`, `pypdf`.
- Code repository: <https://github.com/usfcs-edu/agentik-lensing>.
- Companion Phase 3a reproduction of Huang et al. (2020) lives at `reproductions/hsu-2020/papers/main.t`.

8 Conclusions

The algorithmic core of the Hsu et al. (2025) pair-wise spectroscopic strong-lens search reproduces to within 2–3% on the intermediate counts (13,530 groups / 27,334 spectra versus the published 13,218 / 26,621). All 20 Grade A new candidates explicitly tabulated in Hsu et al. (2025) Table 2 are recovered with median offset $0.91''$ and exact agreement on the published redshift pairs. The $\sim 30:1$ false-positive rate against the published candidate counts is the honest ceiling on an algorithmic-only pipeline that omits visual inspection of spectra (§3.3) and imaging (§4).

The full 28-million-row pipeline runs in ~ 2 minutes on local hardware — considerably faster than the original Phase 2 plan’s 2–4 week estimate, primarily because `spherimatch` turned out to be sublinear in our regime ($t \sim N^{0.63}$).

A Side-by-side comparison with the published figures and tables

For visual verification, the panel below pairs the artifact reproduced in this report (**a, left**) with the corresponding artifact from the published paper (**b, right**). The published panel is reproduced from Hsu et al. (2025) for comparison; the headline agreement is restated in the caption and discussed in full in §4.

References

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- J. Moustakas, J. Buhler, D. Scholte, B. Dey, and A. Khederlarian. FastSpecFit: Fast Spectral Synthesis and Emission-line Fitting of DESI Spectra. 2023. ascl:2308.005.

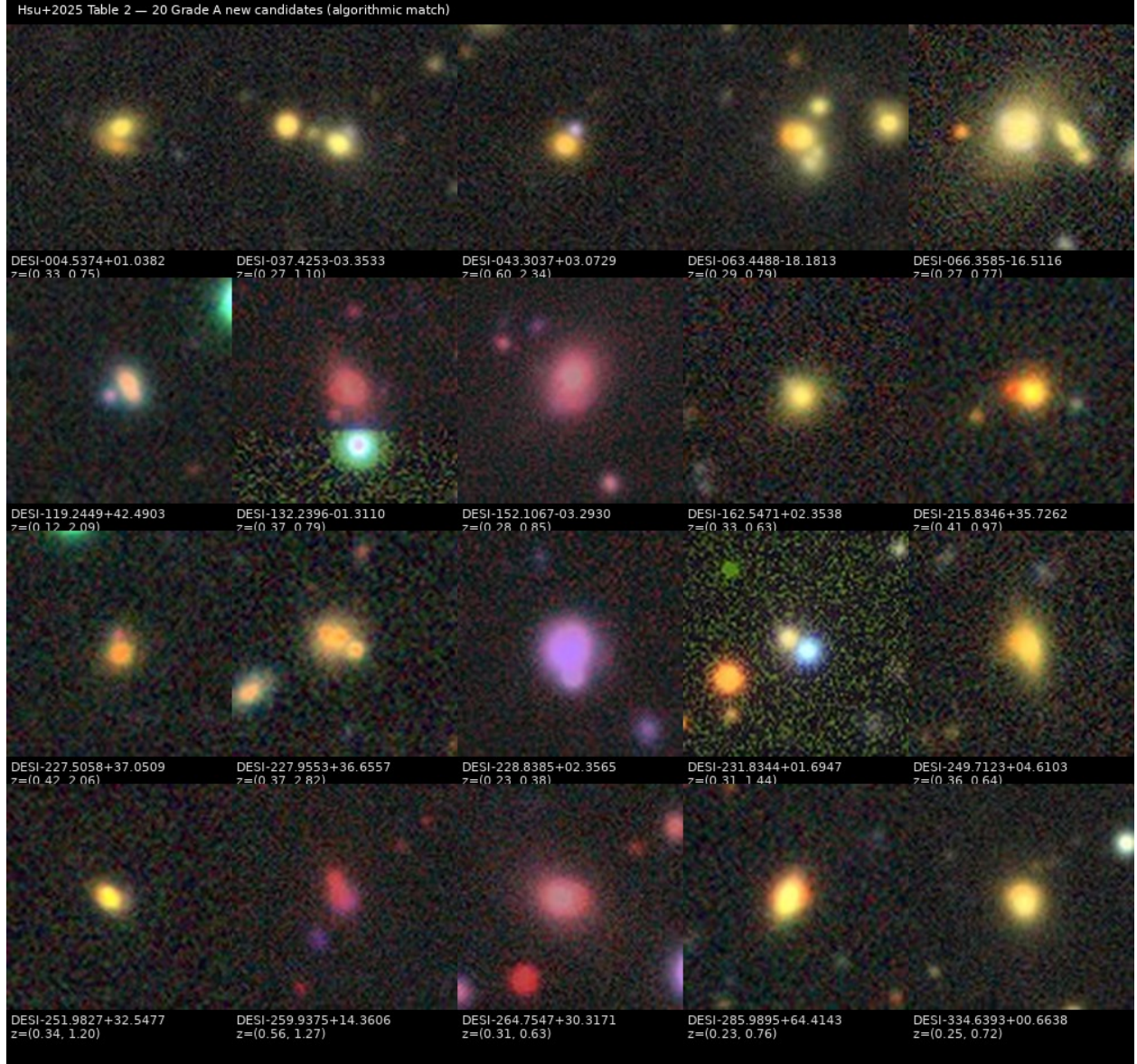
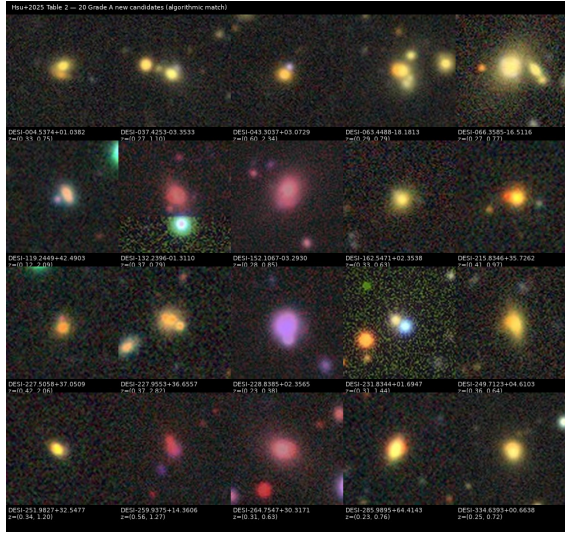
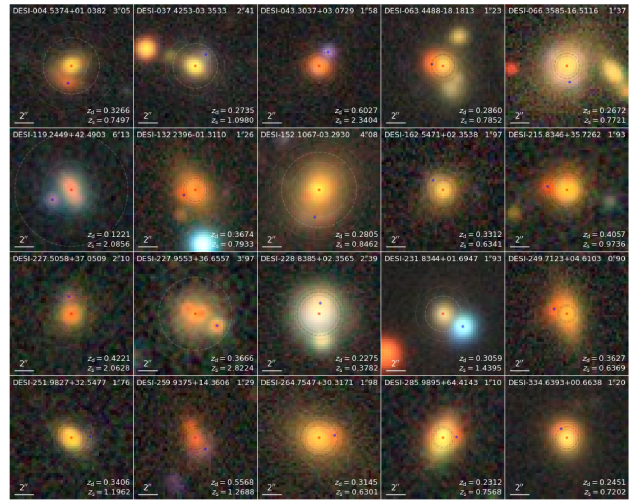


Figure 1: All 20 Hsu+2025 Table 2 Grade A new candidates as recovered in our reproduction. Labels show the published name and the $(z_{\min}, z_{\max})$ redshift pair from our 13,530-group output. Each panel is a 4×5 grz composite at $\approx 25''$ FoV. Many show clear foreground galaxy + nearby background source with arc-like or elongated structure (e.g., DESI-152.1067-03.2930, DESI-228.8385+02.3565, DESI-285.9895+64.4143).



(a) This work (Figure 1)



(b) Hsu et al. (2025), Fig. 3

Figure 2: The 20 Grade A new lens candidates. (a) This work: DR10 *grz* image cutouts at the centroids of the 20 Hsu et al. (2025) Table 2 Grade A systems as recovered by our algorithmic pipeline, labelled with the published name and the $(z_{\min}, z_{\max})$ redshift pair from our 13,530-group output (report Figure 1). (b) The published Figure 3 showing the same 20 candidates, labelled with name and the spectroscopic (z_d, z_s) pair. All 20 of 20 are recovered within $3''$ of our group centroids (median offset $0.91''$, maximum $1.54''$), with the matched group redshifts reproducing the paper's (z_d, z_s) columns to three decimal places. Panel (b) reproduced from Hsu et al. (2025), their Fig. 3, for comparison.